

Kanhaiya Lal

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Career Objective

Computer Science graduate specializing in Machine Learning, with hands-on experience in Python, SQL, Generative AI, RAG, and API development. Experienced in building LLM-based applications, fine-tuning models, and developing machine learning solutions. Seeking to contribute to AI-powered application development, knowledge retrieval, testing, and solution delivery while expanding my expertise in enterprise AI and cloud technologies.

Skills

- **Programming Languages** JavaScript, Java, Python, SQL
- **Computer Vision** OpenCV, Object Detection, Face Recognition, Liveness Detection
- **Machine Learning** PyTorch, TensorFlow, Keras, Scikit-learn, Reinforcement Learning, Deep Learning
- **AI / NLP** LLMs, Hugging Face Transformers, PEFT/QLoRA, RAG, Sentence-Transformers, LangChain
- **Data & Retrieval** Pandas, NumPy, FAISS, PostgreSQL, MySQL, MongoDB
- **Backend & Systems** FastAPI, Flask, Spring Boot, REST APIs, WebSockets
- **Tools/Platforms** Git, GitHub, Docker, Linux, CUDA, AWS, Google Colab, MS Office
- **Soft Skills** Analytical Thinking, Problem Solving, Research Mindset, Critical Thinking, Self-Learning, Collaboration, Adaptability, Technical Communication

Projects

1. Autonomous Stock Trading Agent using Deep Reinforcement Learning (FRIDAY) Jun 2026-Present

- Designed and trained custom DQN and LSTM-DQN agents in PyTorch for the Indian stock market, engineering a 15-dimensional state space (RSI, MACD, Bollinger Bands, Fibonacci levels, Residual Z-Score) with a composite reward function incorporating anti-churn penalties, drawdown minimization, and risk-adjusted returns.
- Built an Evolutionary Arena Tournament system where multiple agent personas (Aggressive, Trend-Following, Mean-Reversion, Neural) compete over rolling evaluation windows; top performers are selected via Sharpe Ratio and Total Return, then reproduced through Gaussian weight mutation (rate=0.08, strength=0.12) to iteratively discover superior trading strategies.
- Implemented a Universal Multi-Asset Trainer using curriculum learning across 50+ NSE/BSE tickers with environment rotation per episode, enabling a single market-agnostic RL model to generalize across stocks and timeframes (1D/1H/15M), with CUDA-accelerated parallel inference powering a Monte Carlo engine that computes VaR, CVaR, and Expected Shortfall across up to 250 simultaneous backtests.
- Integrated an LLM-augmented sentiment pipeline using Google Gemini Flash to score Yahoo Finance RSS headlines into a continuous sentiment signal (−1.0 to +1.0) injected directly into the RL agent's state vector, with a 30-minute cache and multi-key fallback, enabling the agent to incorporate real-time market narrative alongside quantitative indicators.

• **GitHub Link:** <https://github.com/Kanhaiya1610/FRIDAY>

2. JLPT-RAG — Level-Adaptive Japanese Q&A System Sept 2026

- Built an end-to-end Retrieval-Augmented Generation system over a 400-chunk multilingual (English/Japanese) corpus spanning six anime franchises, MEXT/JASSO scholarship documentation, and JLPT vocabulary/grammar references.
 - Designed a query parser that extracts the requested JLPT proficiency level (N5–N1) from natural-language queries, and implemented multilingual semantic retrieval using Sentence-Transformers embeddings over a FAISS vector index.
 - Fine-tuned Llama 3.2 3B with 4-bit QLoRA (PEFT) to generate responses calibrated to the requested JLPT level, conditioning vocabulary and grammar complexity through level-specific system prompts.
 - Implemented a FAISS-based semantic cache with cosine-similarity thresholding, achieving a 100% hit rate on repeated/paraphrased queries and a 157x reduction in response latency (2.54s → 0.02s).
 - Built a baseline-vs-fine-tuned evaluation harness combining a deterministic MeCab-tokenized JLPT-compliance scorer with an LLM-as-judge faithfulness checker, plus a 3-tab Gradio dashboard for live querying, level-by-level comparison, and keyword-vs-semantic retrieval benchmarking.
- **Google Colab Notebook link:** [LLM_NOTEBOOK.ipynb](#)

2. LogiVoice: Smart Logistics Doc & Voice Assistant Jun 2025

- Developed a full-stack web application leveraging AI-powered automation to streamline logistics document workflows, significantly reducing manual effort.
- Implemented multimodal AI capabilities through voice command processing, real-time translation, and document intelligence for PDF extraction, automatic document type detection, and barcode/QR code recognition.
- Designed a user-centric, mobile-friendly dashboard that enabled bulk PDF uploads and incorporated advanced features like annotation tools and searchable/exportable data tables.

- Utilized Natural Language Processing (NLP) and integrated OCR (Tesseract), Google Translate, and speech recognition to facilitate robust multilingual and multimodal input/output.
- Engineered a persistent command and data history with efficient memory management, enhancing system usability and data integrity.
- Built with Python (Flask), JavaScript, Bootstrap, and modern Ai libraries, delivering a high impact solution for logistics automation.
- **GitHub link:** <https://github.com/Kanhaiya1610/LogiVoice>

Internship

Computer Vision & AI Intern | TECHNUM OPUS PVT LTD Oct 2025 – Jan 2026

1. Enterprise Computer Vision & Security Systems (Python, Fast API):

- Engineered two production-grade biometric security solutions: an AI-driven Face Attendance Platform and a Visitor Identity Verification System.
- Implemented robust security protocols including liveness detection (anti-spoofing), multi-frame verification, and government ID matching algorithms to ensure high-accuracy identity assurance.
- Designed a custom Computer Vision pipeline to automate dataset labelling (via Roboflow) and model training, successfully deploying object detection models for Trident (Client) to monitor manufacturing units and security systems.

2. Predictive Analytics & Forecasting:

- Developed predictive models to analyse sales, production, and maintenance datasets, enabling data-driven forecasting of market trends to optimize operational efficiency and reduce costs.

Certificates

- **IBM AI Engineering Professional Certificate** — Coursera · 13 courses Apr 2026
· Deep Learning (Keras, TensorFlow, PyTorch), LLMs, Generative AI, NLP, LangChain, RAG · [Certificate](#)
- **Complete Machine Learning & Data Science Program** July 2024
- **Oracle Cloud Infrastructure 2023 AI Certified Foundations Associate** Jun 2024
- **Attended Workshop on Containers and AI/ML on AWS Cloud by AWS at IIT Roorkee** Mar 2024

Education

- Lovely Professional University** Punjab, India
Bachelor of Technology - Computer Science and Engineering,
Major: Machine Learning | Minor: Japanese Language (N4)
CGPA: 7.58 | Graduated: June 2026
Provisional Degree & Migration Certificate Issued: July 2026 September 2022 – July 2026
- Mohta Public School (CBSE)** Rajgarh, Churu
Senior Secondary April 2020 - March 2021
- Mohta Public School (CBSE)** Rajgarh, Churu
Secondary April 2018 - March 2019

Research Paper (In Preparation)

Multi-Agent Reinforcement Learning with Evolutionary Optimization for Algorithmic Trading on FRIDAY — Kanhaiya Lal 2026

Proposes a novel DRL-based multi-asset trading framework incorporating Kelly Criterion risk sizing, LLM-driven sentiment analysis, and Monte Carlo stochastic back testing with ϵ -exploration noise across simultaneous GPU simulations.

Training

Java Full Stack Training — Cipher School Feb 2025
Built end-to-end web apps using Core Java, Spring Boot, REST APIs, JDBC, MySQL, HTML/CSS/JavaScript · [Certificate](#)

Achievements

- **Reached Semi-finals in a state-level football tournament hosted by a private club in Jaipur** 2024
- **Won Finals in a district-level football tournament hosted by a private club in Churu** 2019

Languages

- **Japanese (日本語):** Conversation Level Proficiency, N4 Equivalent (Reading, Speaking, Writing)
- **Arabic (عربي):** Elementary Level Proficiency (Reading)
- **English:** Advanced, Fluent
- **Hindi (हिंदी):** Native